

b) onto, but not one-to-one.

Solution: $f(n) = \lceil n/2 \rceil$ $f(2) = \lceil 2/2 \rceil = \lceil 1 \rceil = 1$
 $f(1) = \lceil 1/2 \rceil = \lceil 0.5 \rceil = 1$. But $1 \neq 2$.

By Defn. 5, f is not one-to-one.

Let, $f(x) = y$, where $y \in \mathbb{N}$.

$\therefore \lceil x/2 \rceil = y$ If $x = 2y$, then $\lceil x/2 \rceil = \lceil 2y/2 \rceil = y$.

We know $2y \in \mathbb{N}$.

$\therefore \forall y \in \mathbb{N}, \exists x \in \mathbb{N}$ s.t. $f(x) = y$. By Defn. 7, f is onto. $\square$

c) both onto and one-to-one (but different from the identity function.)

Solution: $f(n) = n - 1$

Let, $x, y \in \mathbb{N}$ s.t. $f(x) = f(y) \therefore x - 1 = y - 1 \Rightarrow x = y$.

By Defn. 5, f is one-to-one.

Let, $f(n) = x$, where $x \in \mathbb{N}$. $\therefore n - 1 = x \Rightarrow n = x + 1$. It's easy to see that $x + 1 \in \mathbb{N}$.

$\therefore \forall x \in \mathbb{N}, \exists n \in \mathbb{N}$ s.t. $f(n) = x$. By Defn. 7, f is onto. $\square$

d) neither one-to-one nor onto.

Solution: $f(n) = n^2 - 3n + 2$

We know that $1, 2 \in \mathbb{N}$. $f(1) = 1^2 - 3 \cdot 1 + 2 = 0$ $f(2) = 2^2 - 3 \cdot 2 + 2 = 0$

But $1 \neq 2$. By Defn. 5, f is not one-to-one.

Let, $f(n) = 3$, where $3 \in \mathbb{N}$. $n^2 - 3n + 2 = 3 \Rightarrow n^2 - 3n - 1 = 0 \Rightarrow n = \frac{3 \pm \sqrt{9+4}}{2}$
 $= \frac{3 \pm \sqrt{13}}{2} \notin \mathbb{N}$.

$\therefore$ There does not exist $n \in \mathbb{N}$ s.t. $f(n) = 3$.

By Defn. 7, f is not onto $\square$